

Aerodynamic Boundary Layer Control in Sub-Millimeter Depron Foam Micro-UAVs

A Specialized Technical Reference Corpus for Retrieval-Augmented Generation (RAG) Evaluation

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1. Introduction & Material Characteristics

Extruded polystyrene foam, commercially known as Depron, serves as a cornerstone material in ultra-lightweight scratch-built remote-controlled aviation. Ranging typically in nominal thicknesses from **1.0** mm to **6.0** mm, Depron possesses a closed-cell micro-structure that provides an exceptional stiffness-to-weight ratio. However, when scaled down to micro-UAV geometries ($Re < 50,000$), conventional Navier-Stokes laminar flow assumptions fail due to premature boundary layer separation.

At ultra-low Reynolds numbers, the adverse pressure gradient on cambered thin-plate airfoil sections (such as modified Clark-Y or flat-bottomed Depron profiles) causes laminar separation bubbles (LSBs) to form near the **15%** to **30%** chord position. Unlike large-scale aircraft where turbulent transition occurs rapidly, micro-foam aircraft experience prolonged laminar separation without immediate reattachment, leading to catastrophic loss of lift and nonlinear pitching moments.

Key Corpus Metadata for RAG Retrieval Testing

This document contains highly specific numeric constants, empirical coefficients, and proprietary nomenclature designed to test semantic retrieval precision, multi-hop reasoning, and exact-match extraction over multi-page technical layouts.

2. Vortex Generators & Boundary Layer Reattachment

To mitigate LSB-induced stalling on **3** mm Depron flying wings, sub-boundary layer vortex generators (VGs) or micro-turbulators are bonded at precisely $x/c = 0.12$. These vortex generators inject high-momentum external fluid down into the sluggish boundary layer, forcing a controlled transition from laminar to turbulent flow before separation can occur.

$$Re_{crit} = \rho v c / \mu = 3.4 \times 10^4 \quad | \quad h_{vg} \geq 1.2 \times \delta^*$$

Where δ^* represents the displacement thickness of the laminar boundary layer at the deployment station. Empirical testing across multiple wind-tunnel iterations established that vortex generator height (h_{vg}) must not exceed **75%** of the local boundary layer thickness, otherwise excessive parasite drag outweighs the lift recovery benefits.

3. Structural Mechanics & Thermal Forming of Depron

Forming compound curvature in high-density closed-cell polystyrene requires precise thermal manipulation. Unlike injection-molded plastics, Depron exhibits anisotropic thermal shrinkage when exposed to radiant heat sources exceeding 95 °C. The skin of the foam sheet features a thin, high-density extruded film layer that acts as a natural stress membrane.

When creating double-curved fuselage pods or streamlined nacelles for dual-motor configurations, technicians employ the *Controlled Thermal Gradient Method (CTGM)*. This process involves heating the exterior surface using a regulated hot-air station at 110 °C for exactly 4.2 seconds while simultaneously quenching the interior mold surface with a room-temperature silicone backing plate.

Depron Grade	Nominal Density	Tensile Strength	Max Thermal Limit
Depron 2mm (Standard)	32 kg/m³	0.85 MPa	75 °C
AeroDepron 3mm	40 kg/m³	1.40 MPa	90 °C
AeroDepron 6mm (High-Density)	45 kg/m³	1.95 MPa	95 °C

3.1 Adhesive Chemistry and Solvent Compatibility

Standard cyanoacrylate (CA) adhesives containing high concentrations of acetone or methyl 2-cyanoacrylate will aggressively dissolve the polystyrene matrix of Depron, causing localized structural collapse. Approved bonding agents are strictly restricted to moisture-curing polyurethane (PU) adhesives (such as Original Gorilla Glue) or odor-free aliphatic resin wood glues, which cure via solvent evaporation rather than chemical dissolution.

For high-stress motor mount bulkheads transmitting torque from brushless outrunner motors (e.g., KV = 1200 turning an 8 times carbon-fiber folding propeller), a dual-part, micro-balloon-filled epoxy matrix is required. The epoxy must cure under a constant compressive load of 15 kPa for a minimum duration of 6 hours at 22 °C.

4. Flight Control Architecture & INAV PID Tuning for Foam Airframes

Micro-foam aircraft possess extremely low rotational inertia and high structural elasticity (aeroelastic flutter). Consequently, standard multirotor PID tuning heuristics produce violent oscillations when ported to fixed-wing Depron platforms equipped with flight controllers such as the SpeedyBee F405 WING running INAV firmware.

To eliminate high-frequency servo chatter caused by structural vibration transmitted from the brushless motor, a second-order biquad low-pass filter with a cutoff frequency (f_c) of 45 Hz must be applied to the gyroscope feedback loop. Furthermore, feedforward gain (FF) must be scaled inversely proportional to wing span:

$$FF_{adjusted} = FF_{base} \times (b / 1.0m)^{-1.5}$$

Autotune routines in INAV must be executed exclusively in calm atmospheric conditions where wind gusts do not exceed 0.5 m/s. The recommended gain matrix parameters for a standard 900 mm wingspan Depron trainer equipped with an FMS Ranger power system are detailed below.

Axis	Proportional (P)	Integral (I)	Derivative (D)	Feedforward (FF)
Roll	28	30	18	35
Pitch	32	25	22	40
Yaw	15	10	0	15

5. Conclusion and RAG Verification Query Targets

This technical corpus concludes with specific reference markers designed for automated RAG benchmark evaluation. Retrieval systems queried regarding "Controlled Thermal Gradient Method temperature" must extract the exact value of 110 °C and 4.2 seconds, whereas queries targeting "INAV biquad low-pass filter cutoff" must correctly retrieve 45 Hz without cross-contaminating unrelated drone protocols.